Real-Time Accessibility Dashboards for Higher Education

Beatriz Patatas
fc56950@alunos.fc.ul.pt
LASIGE, Faculdade de Ciências,
Universidade de Lisboa
Lisboa, Portugal

Carlos Duarte
caduarte@edu.ulisboa.pt
LASIGE, Faculdade de Ciências,
Universidade de Lisboa
Lisboa, Portugal

Letícia Seixas Pereira
lspereira@fe.up.pt
INESC TEC, Faculdade de Engenharia,
Universidade do Porto
Porto, Portugal

Abstract

Despite growing commitments to inclusive education, higher education institutions struggle with digital accessibility. Accessibility problems often remain invisible to educators during content creation and difficult for institutions to monitor and manage at scale. We present a Moodle-integrated accessibility system that connects real-time evaluation of PDF materials at upload time with an institutional dashboard that aggregates accessibility metrics across courses and departments. By doing so, the system aims to link everyday teaching practices with higher-level monitoring and decision-making. To understand how educators and coordinators engage with this integrated approach, we conducted a formative user study. Results indicate that while visual feedback and aggregated metrics increase accessibility awareness, a critical tension remains: compliance-oriented technical language hinders educators from translating detection into remediation. These findings suggest that detection and monitoring, while necessary, are not sufficient on their own. Effective accessibility infrastructures must also support meaningful, actionable engagement with accessibility in everyday teaching practice.

CCS Concepts

• **Human-centered computing**;

Keywords

Web Accessibility, Learning Management Systems, Higher Education, Disability, Inclusive Education

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1 Introduction

The importance of inclusive education is increasingly recognized, yet many institutions still struggle to meet accessibility standards fully [5]. Barriers to accessing educational content limit the participation of students with disabilities and negatively affect their academic success and overall university experience [7]. These educational inequalities are reflected in broader societal outcomes: recent data show consistently higher unemployment rates among people with disabilities [6], reinforcing the role of higher education as a key factor in reducing long-term exclusion [6].

A central challenge lies in how digital learning materials are produced and managed. Educators often lack the knowledge, time, and resources required to create accessible content [15], resulting in materials that are difficult or impossible for students with disabilities to use effectively [7]. This issue is particularly acute in Learning Management Systems (LMS). While these platforms act as the primary resource for teaching and learning in higher education [12, 19], they largely function as repositories of inaccessible content allowing instructors to upload inaccessible content without validation or feedback mechanisms [3, 8, 10].

Automated solutions such as Blackboard Ally or the Brickfield Accessibility Toolkit [1, 13] provide mechanisms for institutional content auditing and reporting. Although both are commercial products, Brickfield operates on a "freemium" model that offers basic features at no cost. However, access to their advanced institutional functionalities remains restricted by paid enterprise subscriptions. Beyond these licensing constraints, they share a more fundamental workflow limitation: their reliance on asynchronous, post-hoc scanning. This approach separates the evaluation from the immediate content creation workflow, delaying feedback and reducing the likelihood of prompt remediation by educators.

To address these limitations, this work introduces an open-source, Moodle-integrated system that adopts an institutional-wide perspective. By integrating *synchronous*, real-time accessibility validations at the exact moment a PDF is uploaded, the system connects immediate, actionable feedback for educators with a broader, longitudinal view of accessibility across courses and departments. Furthermore, because the architecture is open-source, it allows other higher education institutions to freely adopt, extend, and customize the tool to fit their specific administrative and pedagogical needs. Ultimately, this approach bridges the gap between everyday teaching practices and proactive institutional decision-making.

To understand the practical impact of this approach, this paper presents a formative evaluation with educators and coordinators. We investigate how they engage with real-time visual feedback and institutional-level metrics. Through this study, we explore what kinds of support and representations are needed for visibility to translate into meaningful action in higher education.

2 Related Work

This section reviews prior research on accessibility in higher education and examines how current tools support, or fail to support, both everyday teaching practices and institutional monitoring.

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2.1 Digital Barriers in Inclusive Education

Inclusive education aims to ensure equal access to learning opportunities for all students, regardless of physical or cognitive abilities [22]. However, the rapid digitization of higher education has often intensified existing inequalities, contributing to a growing digital divide [23].

While previous studies identify a lack of digital skills among educators [5, 15], Sanderson et al. [18] refine this by demonstrating that faculty frequently conflate "digital availability" with "accessibility," erroneously assuming that simply uploading a file to an LMS makes it accessible. This finding is corroborated by findings from Shinohara et al. [21], who demonstrated that even among computing instructors, a demographic expected to be digitally proficient, only 4.5% identified as accessibility experts.

Regarding technical infrastructure, research characterizes the standard LMS as a contributing factor to this "remediation debt." Calvo et al. [4] criticize platforms like Moodle for failing to act as Authoring Tool Accessibility Guidelines (ATAG) compliant environments. Instead of guiding users, these systems function as "passive repositories", accepting inaccessible content without validation. This architectural passivity reinforces what Mack et al. [11] describe as institutional "silos": a structural disconnect where disability services hold the expertise but lack control over content creation, while educators author materials in isolation.

2.2 PDF Accessibility

Among the content types hosted on LMS, PDF documents remain the most widely used format for sharing course materials [26]. However, they often present significant barriers. Common issues include missing alternative text, insufficient contrast, and, inadequate tagging [14].

In an assessment of 200 articles from disability-related journals, Nganji [14] found that only 15.5% were tagged and merely 10.5% included alternative text for images. This lack of tagging creates a fundamental disconnect between what the user sees and what the machine reads. As Shigarov et al. [20] explain, an untagged PDF is merely a collection of loose "text chunks" and visual instructions, lacking any logical reading order. Consequently, trying to reconstruct meaningful structures like tables or paragraphs from these visual cues becomes a complex guessing game for software. Without the explicit structure provided by tags, the content remains inaccessible, making automatic remediation unreliable.

Current remediation strategies impose significant technical burdens. Research found that professional software to evaluate and fix PDF accessibility, such as Adobe Acrobat Pro, is a poorly designed tool and is not all user friendly [16]. Furthermore, the perceived complexity and time cost of these external workflows lead many educators to express unhappiness with existing tools and abdicate responsibility entirely [9]. Thus, a clear gap exists for simplified solutions that can integrate remediation into existing authoring workflows, rather than relying on complex software.

3 System Architecture & Features

To bridge the gap between content creation and institutional monitoring, we designed an accessibility system that integrates directly into the LMS workflow. The system operates via a Moodle plugin that intercepts PDF uploads and transmits them to a Python-based microservice for real-time analysis. Results are immediately stored in Moodle's database, simultaneously populating the instructor's feedback block and the institutional dashboard.

The system targets barriers affecting users reliant on Assistive Technologies (e.g., screen readers). Specifically, it validates document structure to support blind users, and ensures correct language definition to assist users with cognitive disabilities (e.g., dyslexia) using text-to-speech. To support these goals, the analysis engine implements specific W3C PDF Techniques [25] to validate compliance with WCAG 2.2 standards [24]. The system verifies the presence of document tags as a prerequisite (Criterion 1.3.1) and performs the following automated checks: **1) Metadata:** Presence of Document Title (**PDF18**, Criterion 2.4.2), and Default Language (**PDF16**, Criterion 3.1.1); **2) Structure:** Correct semantic tagging of Lists (**PDF21**, Criterion 1.3.1), and Table Headers (**PDF6**, Criterion 1.3.1); and **3) Content:** Presence of alternative text for figures (**PDF1**, Criterion 1.1.1), valid Link annotations (**PDF11**, Criterion 2.4.4), and detection of image-only documents requiring OCR (**PDF7**, Criterion 1.4.5).

To quantify accessibility, the system computes a percentage score for each document based on the ratio of passed tests to the total number of applicable tests. Non-applicable criteria (e.g., table headers in a table-free document) are excluded from the calculation to prevent unfair penalization. At the course level, the overall accessibility metric is computed as the simple average of all individual document scores. Although the system successfully automates the detection of structural tags and metadata, it currently faces two main limitations. First, it evaluates presence rather than semantic quality (e.g., it verifies that an alt-text attribute exists, but cannot determine if the description is actually meaningful to a blind user). Second, the scoring mechanism applies equal weight to all automated tests, without prioritizing critical barriers over minor formatting issues according to WCAG conformance levels (A, AA, and AAA).

3.1 Features for Professors

The system supports instructors by **identifying specific accessibility issues**, **providing contextual support documentation**, and **enabling the tracking of accessibility over time** within the LMS. When a PDF is uploaded, a traffic-light indicator (red, yellow, green) displays its accessibility status directly on the file page. Each check includes a brief explanation of what is being evaluated, and failed checks additionally provide a direct link to the relevant WCAG guidance (see Figure 1). At the course level, a persistent dashboard block summarizes the accessibility status of all materials (see Figure 2). A color-coded progress bar provides an overall view of the course, while a list highlights files with accessibility issues (e.g., "5 of 6 tests failed") and links to detailed reports. A trend graph shows how accessibility evolves over time, allowing instructors to track changes in accessibility status over time at the course level.

3.2 Features for the Institution

While file-level feedback supports individual remediation, institutional change requires visibility at scale. The institutional dashboard

aggregates file-level results into a consolidated view of accessibility across degrees, programs, and courses. Decision-makers can explore the data through hierarchical filters and track how accessibility metrics evolves over time, for example following training initiatives or policy changes. A central table flags programs as *critical*, *warning*, or *good*, based on their overall accessibility score, enabling coordinators to quickly identify where intervention is most needed (see Figure 3). The dashboard also supports benchmarking by highlighting differences across courses and summarizing the most frequent accessibility failures. This overview help institutions prioritize support efforts and target the most recurrent barriers.

4 Preliminary User Study

We conducted a preliminary formative user study to understand how different stakeholders engage with and interpret the accessibility feedback provided by the system. Rather than providing summative validation, the study focuses on how this information shapes both remediation practices and institutional decision-making.

4.1 Participants

The study involved three participants (N=3), recruited through the research team's professional network. A total of four evaluation sessions were conducted, as one participant (P1) took part in two sessions: first in the role of professor and later as a course coordinator.

Participants came from different areas within computing and had varying levels of teaching experience and accessibility knowledge. P1 had 12 years of teaching experience and low accessibility knowledge, being familiar with the concept but not with technical guidelines. P2 had four years of teaching experience and medium accessibility knowledge, including familiarity with some accessibility terminology. P3 had one year of teaching experience and low accessibility knowledge.

4.2 Study Protocol

Before starting the study, participants were informed about the research goals and provided informed consent. A dedicated testing environment was prepared by configuring a local Moodle instance with sample courses populated with a curated mix of compliant and non-compliant PDF files. This setup allowed the system to present realistic accessibility states during the sessions. The study followed a think-aloud protocol, in which participants were asked to verbalize their thought process while performing a series of predefined tasks. Each session lasted approximately 60 minutes and was conducted in person. All participants completed the tasks in the **professor role**, while one participant (P1) additionally performed tasks in the **course coordinator role**.

In the **professor role**, participants uploaded PDFs files (e.g., scanned chapters and untagged syllabuses), received real-time accessibility feedback and attempted to interpret the reported issues using the provided guidance to remediate at least one accessibility problem. In the **coordinator role**, P1 interacted with the institutional dashboard to identify programs flagged as *critical* and to explore accessibility trends at the institutional level.

All sessions were recorded with participants' permission. After completing the tasks, a semi-structured interview was conducted to capture participants' reflections on the system and its perceived usefulness.

4.3 Data Analysis

Qualitative data, including observation notes and audio-visual recordings, was analyzed using an inductive thematic analysis approach following Braun and Clarke [2].

The dataset was iteratively coded to identify recurring patterns and critical incidents related to system use and interpretation of feedback. Codes were then grouped into higher-level themes reflecting usability challenges and the understandability of the system's guidance. Observations were triangulated with post-task interview responses to align reported perceptions with observed behavior.

5 Preliminary Results: User Study

5.1 Key Findings

The analysis conducted highlighted three primary themes concerning how accessibility feedback was understood, acted upon, and leveraged at different institutional levels.

5.1.1 What you see is not what you get. The study revealed a clear mismatch between how educators perceive document structure and how accessibility tools evaluate it. Participants tended to equate visual formatting with accessibility, overlooking the role of underlying semantic structure.

This was most evident with P3, who initially expressed high confidence in their materials, rating the importance of accessibility as "5/5" while admitting that they never actively checked it. When the system flagged a "Table Header Check Failed," P3 justified the result by referring exclusively to visual formatting choices rather than semantic markup:

> *"I never do that [tag headers]. I just go there, increase the font size, and put it in bold."* – (P3)

A similar lack of awareness emerged around document metadata. All participants expressed surprise when the system flagged a "Missing Title" error, revealing that they were unaware that PDFs rely on internal document properties that are distinct from the visible filename. As P1 remarked:

> *"I didn't know it checked the internal properties... for someone starting from zero, this is not obvious."* – (P1)

5.1.2 A barrier to breaking barriers. When attempting to remediate the issues flagged by the system, participants struggled to translate detection into concrete action, even when motivated to fix the problem. Difficulties emerged both in locating the correct fix and engaging with the support materials provided.

P3 initially ignored the help links, assuming that correcting issues such as missing document properties would be straightforward within the authoring software. This assumption quickly proved incorrect as after a brief period of unsuccessful exploration, P3 acknowledged that the solution was not obvious and that external guidance would be necessary to proceed.

By contrast, P1 immediately accessed the provided help documentation but encountered a different barrier. When redirected to WCAG-based guidelines, they disengaged from the resource due to

its density and technical complexity, indicating that the information was not actionable in practice:

> *"It was not very useful because it is too technical and I don't want to read all that... I just want a small summary."* – (P1)

5.1.3 Empowering Institutional Oversight. The evaluation with P1 in the coordinator role highlighted how the dashboard re-frames accessibility from a hidden technical concern into an explicit managerial responsibility. Having access to aggregated, program-level data removed what P1 described as the "ignorance gap", making inaction visible and harder to justify:

> *"Without this, I do nothing. With this I would email the professors... It is so easy to generate a report that doing nothing would be negligent."* – (P1)

Beyond awareness, the dashboard enabled concrete institutional action. P1 emphasized that the exported reports were sufficiently clear and robust to be shared with higher-level administration (e.g., Rectorate) as part of formal quality assurance processes. The traffic light visualization further supported rapid prioritization, allowing coordinators to immediately identify critical courses and direct support when it was most needed.

6 Discussion

The findings highlight that accessibility challenges in LMS environments are not primarily caused by a lack of technical tooling, but a persistent misalignment between everyday authoring practices and technical criteria used to assess accessibility. Educators tend to rely on visual formatting conventions that appear structurally meaningful to them, yet remain invisible to assistive technologies. As a result, accessibility issues in formats such as PDF are systematically overlooked during content creation.

This gap suggests that effective accessibility support must go beyond automated detection. Simply reporting errors is insufficient if users cannot understand how their visual choices translate or fail to translate into machine-readable structure. The results indicate that tools should make this disconnect explicit, helping educators grasp why familiar practices (e.g., font size or bold text) do not provide the semantic information required by accessibility standards.

The remediation task further reveals a second tension: while participants were willing to address reported issues, existing forms of technical guidance were perceived as too complex to support action. The rejection of WCAG-based documentation underscores a well-known limitation of compliance-oriented approaches, where standards are necessary but not directly usable by non-experts. This suggests that accessibility infrastructures need intermediary layers that translate standards into concise, contextualized guidance embedded within authoring workflows.

At the institutional level, the study shows that aggregating accessibility data changes how accessibility is framed and acted upon. By presenting accessibility as a collective and measurable property of courses and programs, the dashboard shifts responsibility from isolated individual effort to coordinated institutional action. This macro-level visibility enables prioritization, accountability, and targeted support, reinforcing the role of accessibility as an organizational concern rather than a purely technical one.

7 Future Work

Future work will focus on validating the system in a real-world educational setting. In particular, we plan to deploy the tool in active university courses over a full semester, allowing us to observe how instructors engage with accessibility feedback under everyday time constraints and competing teaching demands. This deployment will enable a deeper assessment of whether the system supports sustained changes in authoring practices beyond controlled evaluation settings.

Beyond evaluating the system's long-term impact, we also plan to refine its feedback mechanisms to address the observed struggles with technical terminology. To overcome the bureaucratic and dense nature of formal WCAG documentation, future iterations will focus on translating complex standards into dynamic, actionable, and plain-language guidance. Integrating simplified and visual resources, such as the WCAG Guide developed by Sales [17], has the potential to transform the tool from a simple compliance checker into an intuitive pedagogical aid.

Building on this validation, future iterations of the system will extend the monitoring pipeline to additional content formats commonly used in higher education, including Microsoft Word, PowerPoint, LaTeX, and native LMS course pages, with the goal of supporting accessibility across the full range of instructional materials.

8 Conclusion

This paper presented the design and formative evaluation of a dual-layer accessibility monitoring system integrated into an LMS. By embedding real-time feedback into the moment of content upload and complementing it with aggregated institutional views, the system connects individual authoring practices with broader organizational oversight.

Taken together, the results suggest that effective accessibility infrastructures in higher education must combine detection, translation, and oversight. Addressing the digital divide requires not only identifying technical barriers, but also supporting the people and institutions responsible for removing them.

References

[1] Blackboard Inc. 2025. Blackboard Ally – Accessible content is better content. https://ally.ac/. Accessed: August 19, 2025.

[2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.

[3] Elisabeth Brito and Goncalo Paiva Dias. 2020. LMS accessibility for students with disabilities: The experts' opinions. *2022 17th Iberian Conference on Information Systems and Technologies (CISTI)* (6 2020), 1–5. doi:10.23919/cisti49556.2020.9141046

[4] Rocío Calvo, Ana Iglesias, and Lourdes Moreno. 2014. Accessibility barriers for users of screen readers in the Moodle learning content management system. *Universal access in the information society* 13, 3 (2014), 315–327.

[5] Weiqin Chen. 2021. Students with disabilities and digital accessibility in higher education under COVID-19. In *Proceedings of the 29th International Conference on Computers in Education*. Asia-Pacific Society for Computers in Education.

[6] Eurostat. 2025. Disability statistics - unemployment and people outside the labour force. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Disability_statistics_-_unemployment_and_people_outside_the_labour_force Data extracted in October 2025.

[7] José María Fernández-Batanero, Marta Montenegro-Rueda, and José Fernández-Cerero. 2022. Access and Participation of Students with Disabilities: The Challenge for Higher Education. *International Journal of Environmental Research and Public Health* 19, 19 (2022). doi:10.3390/ijerph191911918

[8] Ginikachi Prisca Ifenatuora, Olanrewaju Awoyemi, and Fadeke Adeola Atobatele. 2023. Systematic review of accessibility compliance in LMS platforms for inclusive online learning. *International Journal of Multidisciplinary Research and Growth Evaluation* 4, 2 (2023), 785–791.

[9] A Jembu Rajkumar, JB Jordan, and J Lazar. 2020. Improving PDF accessibility tools for content developers: looking towards the future. In *Cambridge Workshop on Universal Access and Assistive Technology*. Springer, 173–181.

[10] Jeffrey P Kontio Jr. 2025. From reactive to proactive: Leveraging Learning Management Systems for a more accessible learning experience. In *Proceedings of the 2025 ACM SIGUCCS Annual Conference*. 1–4.

[11] Kelly Avery Mack, Natasha A Sidik, Aashaka Desai, Emma J McDonnell, Kunal Mehta, Christina Zhang, and Jennifer Mankoff. 2023. Maintaining the Accessibility Ecosystem: a Multi-Stakeholder Analysis of Accessibility in Higher Education. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility* (New York, NY, USA) *(ASSETS '23)*. Association for Computing Machinery, New York, NY, USA, Article 100, 6 pages. doi:10.1145/3597638.3614547

[12] Javier Mella-Norambuena, Andres Chiappe, and Maria Graciela Badilla-Quintana. 2025. Theoretical and empirical models underlying the teaching use of LMS platforms in higher education: a systematic review. *Journal of Computers in Education* 12, 3 (2025), 997–1023.

[13] Moodle.org. 2025. Accessibility Toolkit – MoodleDocs. https://docs.moodle.org/405/en/Accessibility_Toolkit. Accessed: August 19, 2025.

[14] Julius T. Nganji. 2018. An assessment of the accessibility of PDF versions of selected journal articles published in a WCAG 2.0 era (2014–2018). *Learned Publishing* 31, 4 (2018), 391–401. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/leap.1197 doi:10.1002/leap.1197

[15] Andrea Osuna-Juárez and Nuria González-Castellano. 2024. Understanding professors' and students with disabilities' perceptions of inclusive higher education: a systematic literature review. *European Journal of Special Needs Education* 39, 5 (2024), 774–788. arXiv:https://doi.org/10.1080/08856257.2023.2294238 doi:10.1080/08856257.2023.2294238

[16] Aravind Jembu Rajkumar, Jonathan Lazar, J Bern Jordan, Alireza Darvishy, and Hans-Peter Hutter. 2020. PDF accessibility of research papers: What tools are needed for assessment and remediation?. In *HICSS*. 1–10.

[17] Marcelo Sales. 2018. Guia WCAG: Guia de consulta rápida. https://guia-wcag.com/ Accessed: 2026-02-27.

[18] Norun Christine Sanderson, Siri Kessel, and Weiqin Chen. 2022. What do faculty members know about universal design and digital accessibility? A qualitative study in computer science and engineering disciplines. *Universal Access in the Information Society* 21, 2 (2022), 351–365.

[19] Mahammad Sharifov, Samaya Safikhanova, and Abdulsalam Mustafa. 2021. Review of prevailing trends barriers and future perspectives of learning management systems (lmss) in higher education institutions. *International Journal of Education and Development using Information and Communication Technology* 17, 3 (2021), 207–216.

[20] Alexey Shigarov, Andrey Mikhailov, and Andrey Altaev. 2016. Configurable table structure recognition in untagged pdf documents. In *Proceedings of the 2016 ACM symposium on document engineering*. 119–122.

[21] Kristen Shinohara, Saba Kawas, Amy J Ko, and Richard E Ladner. 2018. Who teaches accessibility? A survey of US computing faculty. In *Proceedings of the 49th ACM Technical Symposium on Computer Science Education*. 197–202.

[22] Roger Slee. 2018. Defining the scope of inclusive education. (2018).

[23] Viktoriia V Sydorenko, Inna O Akhnovska, Sergiy Smirnov, Igor Verbovskyi, and Olha V Melnychuk. 2024. Improvement of higher education: how to bridge the digital divide during the transformation? *Journal of Education and Learning (EduLearn)* 18, 3 (2024), 1001–1014.

[24] W3C Web Accessibility Initiative (WAI). 2025. WCAG 2 Overview – Web Content Accessibility Guidelines. https://www.w3.org/WAI/standards-guidelines/wcag/. Accessed: August 19, 2025.

[25] World Wide Web Consortium (W3C). 2016. PDF Techniques for WCAG 2.0 and later. https://www.w3.org/WAI/WCAG21/Techniques/#pdf Accessed: 2026-02-26.

[26] Ahmet Çakir. 2016. Usability and accessibility of portable document format. *Behaviour & Information Technology* 35, 4 (2016), 324–334. arXiv:https://doi.org/10.1080/0144929X.2016.1159049 doi:10.1080/0144929X.2016.1159049

A Appendix

This appendix provides visual examples of the system's integration within the Moodle LMS, demonstrating both the instructor's immediate feedback and the institutional monitoring dashboard.

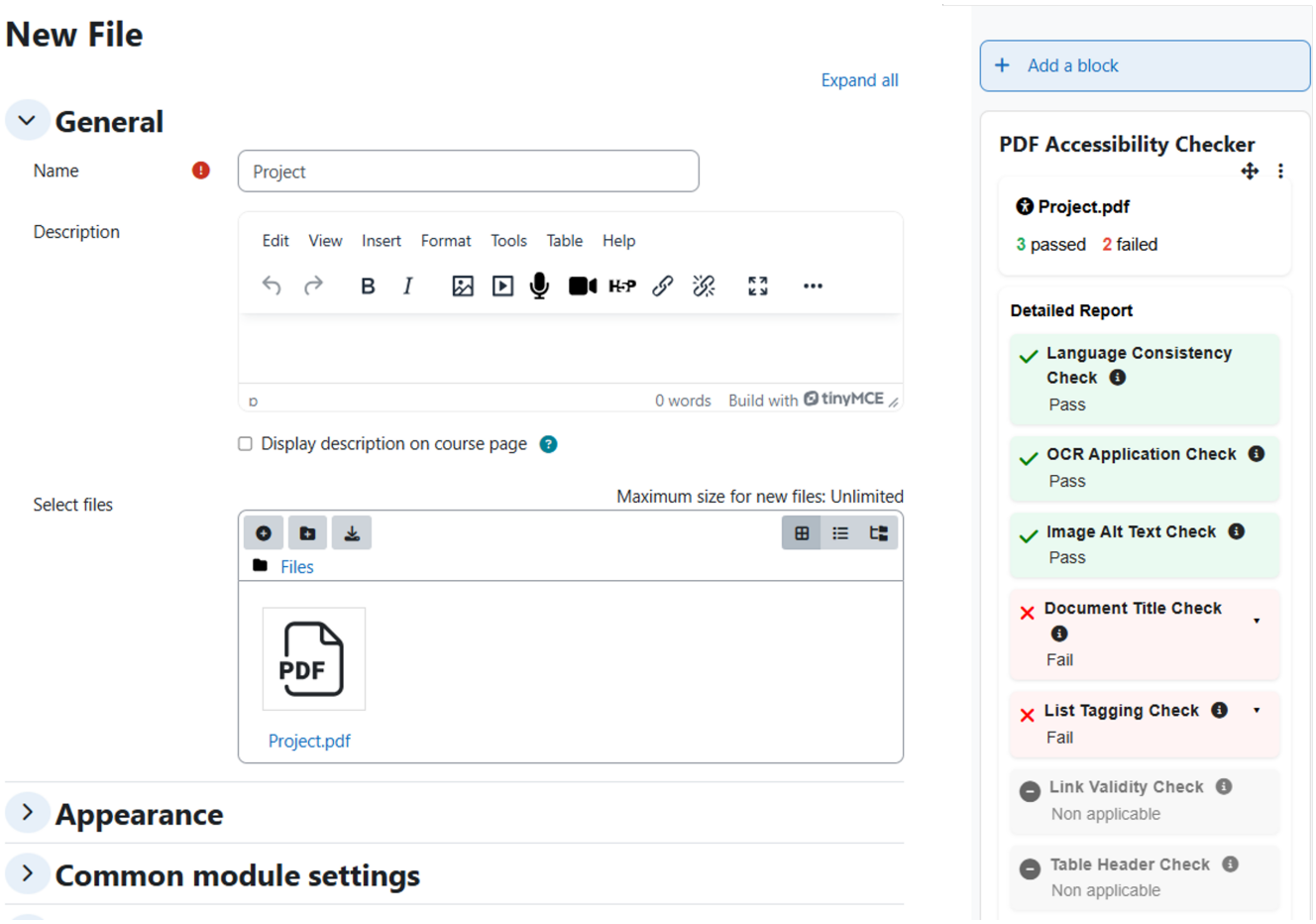

Figure 1: Instructor view: Real-time accessibility feedback block displayed immediately after a PDF is uploaded in Moodle, highlighting passed checks and errors

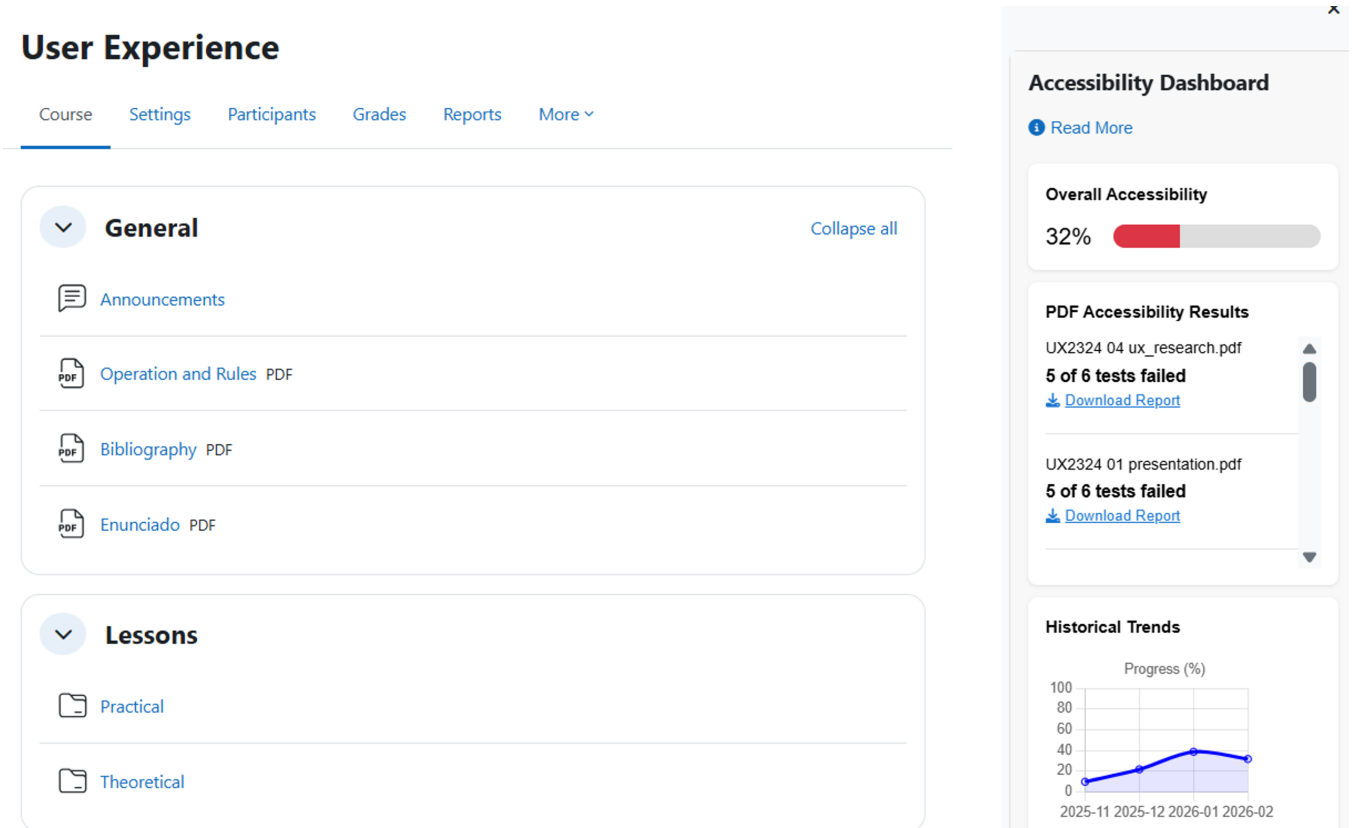

Figure 2: Course-level Accessibility Dashboard: An integrated Moodle block providing the instructor with the course's overall accessibility score, a breakdown of individual PDF results with downloadable reports, and a historical trend chart.

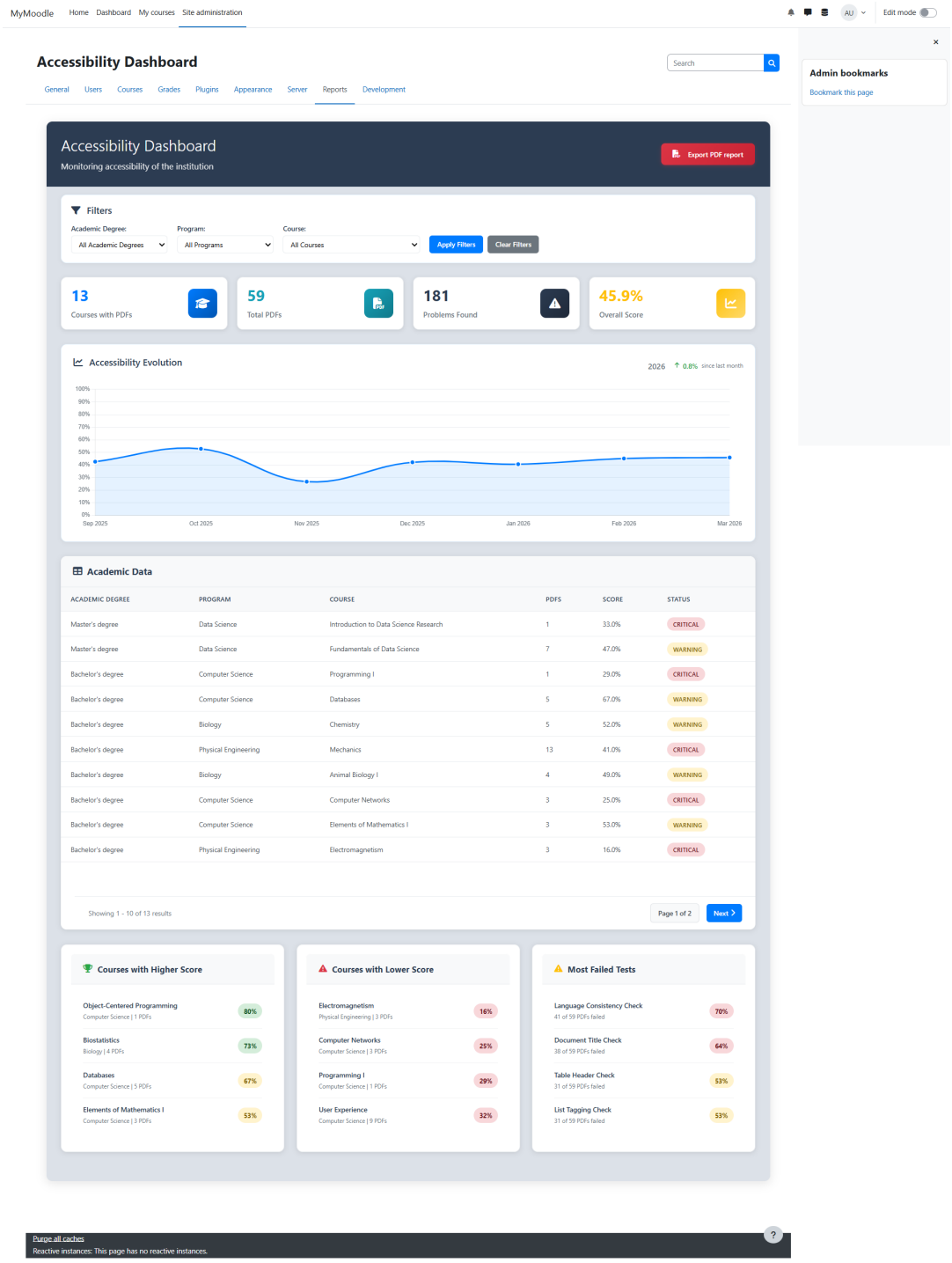

Figure 3: The Institutional Dashboard. Top: Filters to select specific Degrees, Programs or Courses. Middle (Left): "Accessibility Evolution" graph showing progress over time. Middle (Right): Course list highlighting "Critical" and "Warning" statuses. Bottom: Summary of highest and lowest scoring courses, alongside a list of the most frequent accessibility errors.